STUDYRAG â€” PILOT DEPLOYMENT PLAN

Offline-Capable, Voice-Driven AI Teaching at Scale | 1,000+ Users | 20-Week Academic Pilot

1. Primary Objectives and Structure of the Pilot

The central question this pilot is designed to answer is a practical one: can a voice-driven, offline-capable AI tutoring platform produce meaningful gains in comprehension, engagement, and long-term retention for students who are routinely excluded from modern educational technology â€” not because they lack motivation or ability, but because they lack a reliable internet connection or because existing tools were not designed with their cognitive or physical needs in mind?

StudyRAG's architecture treats low-bandwidth environments as a first-class constraint rather than an afterthought. Lesson content is pre-cached locally, AI tutoring sessions run entirely on-device, and progress syncs silently during brief connectivity windows without interrupting the student's workflow. The pilot exists to stress-test this architecture at scale â€” 1,000 active users across diverse contexts â€” and to generate the kind of longitudinal evidence that institutional partners and funders need before committing to permanent adoption.

Four-Phase Structure

- **Phase 1 â€” Preparation and Onboarding (Weeks 1â€“4):** During this phase we finalize institutional agreements, build and quality-check the localized curriculum cache for each deployment region, and run train-the-trainer workshops with educators and NGO field staff. The goal is that by Day 1 of deployment, every supporting adult has already used the platform themselves and can answer a student's first-hour questions without escalating to our team.
- **Phase 2 â€” Deployment and Initial Use (Weeks 5â€“8):** The application is distributed to the full cohort of 1,200 enrolled students. Usage telemetry â€” session length, concept completion rates, voice-query frequency, and offline vs. online session ratios â€” is collected continuously and auto-synced to our analytics backend whenever a device connects. No student action is required for data to upload.
- **Phase 3 â€” Active Monitoring and Iteration (Weeks 9â€“16):** This is the longest and most operationally intensive phase. We conduct bi-weekly reviews of aggregated usage data, push Over-the-Air (OTA) content updates and UI refinements through high-speed sync nodes, and run short accessibility sprints responding to feedback from students with disabilities. Mid-pilot surveys are distributed at Week 10 and Week 14 to capture qualitative experience data alongside the quantitative telemetry.
- **Phase 4 â€” Evaluation and Reporting (Weeks 17â€“20):** The final phase centers on summative learning assessments benchmarked against each institution's existing academic standards, a comprehensive user satisfaction survey, and structured debrief sessions with all partner organizations. A full impact report â€” including disaggregated outcomes by user segment â€” is delivered to all stakeholders within 30 days of the pilot's close.

2. Duration of the Pilot

The pilot runs for **20 weeks (five calendar months)**, deliberately aligned with a standard academic semester so that participation can be integrated into students' existing coursework rather than added on top of it. A 20-week window is long enough to observe whether initial engagement persists or declines after the novelty period â€” typically Weeks 3 through 6 â€” and long enough to capture meaningful learning-outcome data from the summative assessments administered in the final phase. Pilots shorter than 12 weeks routinely overstate engagement metrics because they end before habit decay sets in; this design explicitly accounts for that risk.

3. Participating User Groups

We are enrolling **1,200 students** in total, building in a 20% buffer to ensure that the active user base remains at or above 1,000 throughout the pilot even after anticipated attrition from course drops, device failures, or relocation. The cohort is divided into three segments, each chosen to generate distinct and complementary evidence.

Low-Resource and Marginalized Students (60% â€” approx. 720 students)

This segment is the primary beneficiary group and the core reason StudyRAG exists. Students are drawn from low-income rural and peri-urban households where internet access is intermittent, prohibitively expensive, or both. Many of these students have previously been excluded from AI-assisted learning tools not because those tools wouldn't help them, but because those tools require a stable broadband connection to function at all. StudyRAG's offline-first architecture removes that dependency entirely. Recruitment for this segment happens primarily through our NGO partners who already operate in these communities and have earned the trust required to make participation feel safe rather than extractive.

Students with Learning Disabilities (20% â€” approx. 240 students)

This segment focuses specifically on students with Dyslexia and ADHD, two conditions that are frequently underserved by text-heavy educational interfaces. StudyRAG's voice-first interaction model, adjustable pacing, short-form concept chunking, and WCAG 2.1 AA-compliant UI are designed to minimize cognitive and reading load. Participants in this segment are identified through partnerships with university and school-district Disability Services offices, who also co-design the accessibility-specific feedback instruments used during Phases 3 and 4. Outcomes for this segment are tracked separately, with particular attention to session completion rates, average session length, and self-reported comprehension confidence.

General Cohort â€” Control Baseline (20% â€” approx. 240 students)

This segment is drawn from standard-resource environments â€” students with reliable connectivity and no documented learning disabilities â€” and serves as a comparative baseline. Including this group allows us to determine whether StudyRAG's engagement and outcome improvements are specific to the populations it was designed for, or whether they generalize more broadly. It also helps institutional partners contextualize results against their existing student population averages.

4. Potential Partnerships

No single organization has the distribution reach, community trust, technical infrastructure, and accessibility expertise required to run this pilot well. Our partnership model deliberately combines four complementary actor types:

- **Universities and School Districts (Public/System-Level):** Institutional partners provide the formal channels through which software is distributed to students, integrate pilot participation into existing course credit or extracurricular structures, share anonymized academic performance baselines against which we can measure learning gains, and give the pilot the institutional legitimacy that standalone app deployments lack. Target partners include at least two regional universities and one school district with an established rural or peri-urban student population.
- **Educational Equity NGOs:** Ground-level NGO partners handle recruitment and onboarding for the low-resource segment, where institutional channels alone are insufficient. They conduct localized onboarding workshops in the communities' primary languages, distribute devices where students lack access to their own hardware, and serve as the primary qualitative feedback channel â€” gathering observations, anecdotes, and concerns that usage dashboards are structurally unable to surface.
- **Telecommunication Providers:** We are in active exploratory conversations with regional telecom operators about zero-rating arrangements that would exempt StudyRAG's sync and cache-download traffic from data charges for students on low-income mobile plans. Even a partial zero-rating agreement substantially lowers the barrier to connectivity for the low-resource segment and strengthens the case for long-term scalability.

- **University Accessibility and Disability Services Offices:** These offices bring specialized expertise in assistive technology, neurodivergent learning patterns, and ethical research design with disabled participants. They co-design the feedback rubrics for the disability segment, help ensure that recruitment processes are genuinely voluntary and non-coercive, and flag any UI regressions introduced through OTA updates that might disproportionately affect screen-reader or switch-access users.

5. Responsibilities of Each Partner

StudyRAG (Core Team)

We are responsible for the software itself â€” including a 99.9% uptime SLA for all server-side components, end-to-end encryption of student data, compliance with applicable data privacy regulations (FERPA, GDPR as applicable), and the delivery of aggregated, anonymized analytics reports to institutional partners on a bi-weekly basis. We also lead all UX iteration sprints, manage the OTA update pipeline, and provide a dedicated technical support channel with a guaranteed 24-hour response time for critical issues.

Educational Institutions

Partner institutions are responsible for formally integrating pilot participation into their student engagement structures â€” through course credit, extra credit, or official extracurricular recognition â€ to ensure that participation is incentivized appropriately. They are also responsible for providing anonymized academic performance baseline data (prior-semester grades, standardized test scores where available) that allows us to control for pre-existing differences between user segments when analyzing outcomes.

NGO Partners

NGO partners serve as the human infrastructure of the low-resource segment. They are responsible for community recruitment, culturally appropriate onboarding, device distribution logistics, and ongoing qualitative data collection through structured field observations and informal interviews. They are not responsible for technical troubleshooting, but they receive training to diagnose and escalate the most common device and software issues without requiring a student to contact our team directly.

Accessibility and Disability Services Offices

These offices take shared ownership of the research instruments used with the disability segment. They review and approve all surveys and interview guides before deployment, participate in the mid-pilot data interpretation sessions, and produce a separate accessibility findings memo as part of the Phase 4 reporting package. Their involvement ensures that findings from this segment are interpreted through an appropriate lens rather than treated as a simple subset of the general results.

6. Long-Term Outcomes and Persistence Pathways

This pilot is not a one-time intervention. It is designed from the first week to build infrastructure, relationships, and institutional habits that outlast the 20-week window.

- **LMS Integration:** The most significant long-term outcome target is direct integration of StudyRAG's AI tutoring layer into the Learning Management Systems already in use by institutional partners â€” Moodle, Canvas, or Blackboard depending on the institution. Rather than requiring students to switch between platforms, the goal is that by Month 6 post-pilot, StudyRAG's voice tutoring and adaptive quiz features appear natively within the LMS environment students already use daily. This removes the adoption friction that kills most third-party educational tools after a pilot ends.
- **Data-Driven Curriculum Support:** Throughout the pilot, the analytics engine logs which concepts trigger the most repeated queries, where students disengage mid-session, and which voice explanations most reliably produce correct responses on follow-up questions. This data is returned to institutional partners in an

actionable format â€” not as raw logs but as ranked lists of curriculum gaps with suggested remediation strategies. The intent is that this feedback loop becomes a standing component of each institution's annual curriculum review process, independent of whether StudyRAG remains in use.

- **Sustained Offline Access Infrastructure:** The low-bandwidth caching and sync architecture built for this pilot is not proprietary in a way that locks institutions to StudyRAG. We commit to publishing the technical specifications for the sync protocol under an open license post-pilot, allowing partner institutions and other developers to apply the same offline-first model to other educational tools. This establishes a durable infrastructure precedent rather than a one-time deployment.
- **Policy and Advocacy Pathway:** The disaggregated outcome data generated by this pilot â€” particularly for the low-resource and disability segments â€” is intended to feed directly into policy conversations at the regional education authority level about what accessibility and connectivity standards should be required of publicly funded EdTech tools. NGO partners and Disability Services offices will co-author a public policy brief drawing on pilot findings, delivered no later than 60 days after the pilot closes.